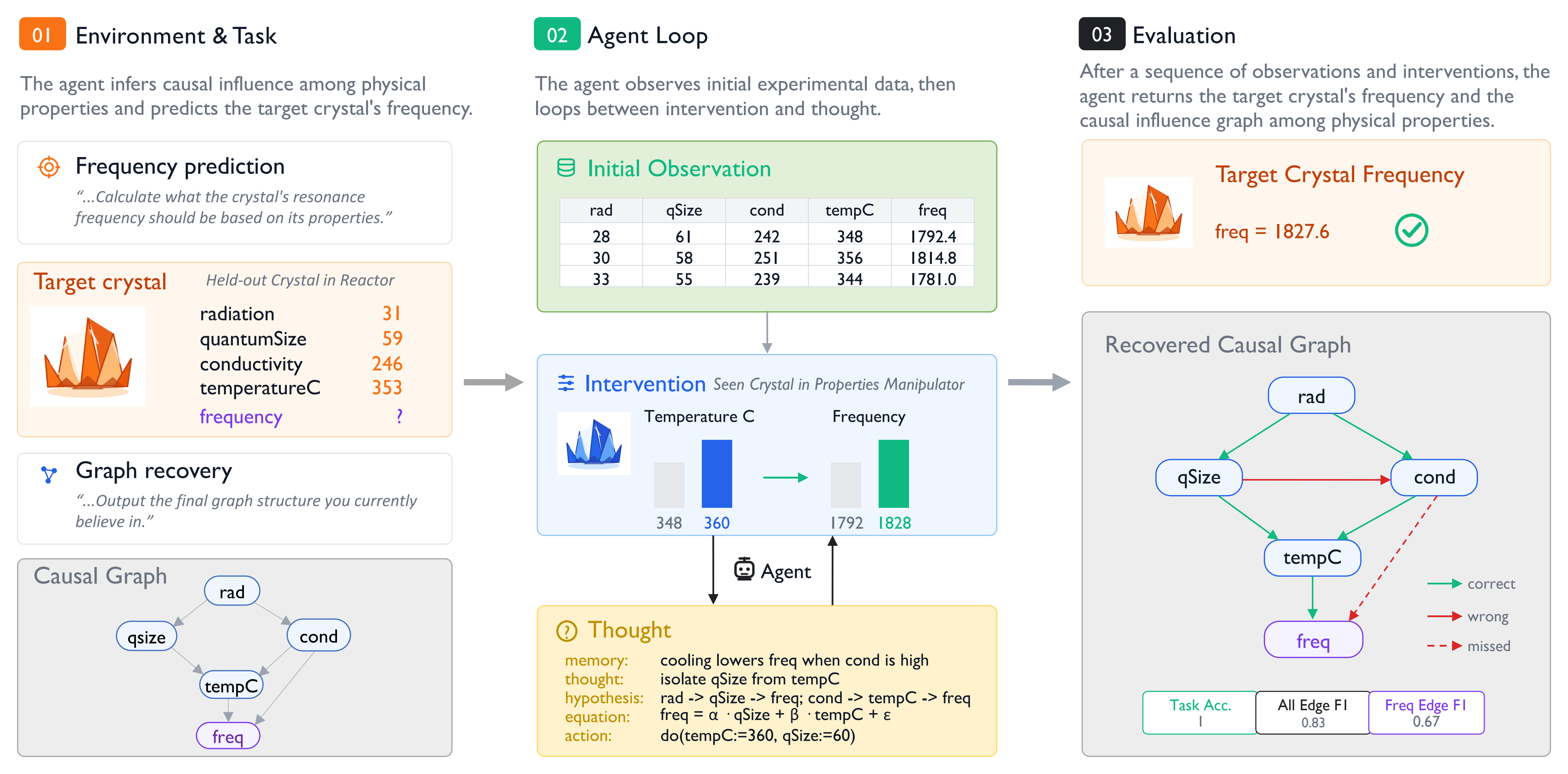
01 Environment & Task
The agent infers causal influence among physical properties and predicts the target crystal's frequency.
Frequency prediction
"...Calculate what the crystal's resonance frequency should be based on its properties."
Target crystal
Held-out Crystal in Reactor
radiation 31
quantumSize 59
conductivity 246
temperatureC 353
frequency ?
Graph recovery
"...Output the final graph structure you currently believe in."
Causal Graph
rad
qsize
cond
tempC
freq
02 Agent Loop
The agent observes initial experimental data, then loops between intervention and thought.
Initial Observation
rad qSize cond tempC freq
28 61 242 348 1792.4
30 58 251 356 1814.8
33 55 239 344 1781.0
Intervention
Seen Crystal in Properties Manipulator
Temperature C
Frequency
348 360
1792 1828
Agent
Thought
memory: cooling lowers freq when cond is high
thought: isolate qSize from tempC
hypothesis: rad -> qSize -> freq; cond -> tempC -> freq
equation: freq = α · qSize + β · tempC + ε
action: do(tempC:=360, qSize:=60)
03 Evaluation
After a sequence of observations and interventions, the agent returns the target crystal's frequency and the causal influence graph among physical properties.
Target Crystal Frequency
freq = 1827.6
Recovered Causal Graph
rad
qSize
cond
tempC
freq
correct
wrong
missed
Task Acc.
1
All Edge F1
0.83
Freq Edge F1
0.67